

True or false? (unless otherwise specified.)

Answer grid

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	B
True	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
False	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

1. Let $x_0 < x_1 < x_2$. If $p, q \in \mathcal{P}_2$ satisfy

$$p(x_i) = y_i, \quad q(x_i) = z_i, \quad i = 0, 1, 2,$$

then $p + q$ interpolates the data $(x_i, y_i + z_i)$ for $i = 0, 1, 2$.

2. For all $x_0, x_1, y_0, y_1 \in \mathbb{R}$ with $x_0 \neq x_1$, there exists a unique $p \in \mathcal{P}_2$ such that

$$p(x_0) = y_0, \quad p(x_1) = y_1.$$

3. If $x_0 < \dots < x_n$ and $\ell_i \in \mathcal{P}_n$ is the i th Lagrange basis polynomial, then by definition

$$\ell_i(x_j) = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases} \quad i, j = 0, \dots, n.$$

4. If $p, q \in \mathcal{P}_n$ and $p(x_i) = q(x_i)$ for $i = 0, \dots, n$, where $x_0 < \dots < x_n$, then $p = q$.

5. If $f \in \mathcal{P}_d$ and $p_n \in \mathcal{P}_n$ is its interpolant at the $n + 1$ distinct nodes $x_0, \dots, x_n$, with $n \geq d$, then $p_n = f$.

6. **Interpolation error.** Let $f \in C^{n+3}[a, b]$, let $a \leq x_0 < \dots < x_n \leq b$, and let $p_n \in \mathcal{P}_n$ interpolate f at $x_0, \dots, x_n$. Then, for every $x \in [a, b]$, there exists $\xi_x \in (a, b)$ such that

$$f(x) - p_n(x) = \frac{f^{(n+3)}(\xi_x)}{(n+3)!} \prod_{i=0}^n (x - x_i).$$

7. **Interpolation error.** Let $u(x) = x^4 + 2x^3 + 3x^2 + 4$ and let $p_3 \in \mathcal{P}_3$ be its interpolant at the points 1, 2, 3, 4. Then

$$u(x) - p_3(x) = (x-1)(x-2)(x-3)(x-4), \quad \forall x \in \mathbb{R}.$$

8. **Convergence.** For every function $f \in C[-1, 1]$, if p_n is the polynomial interpolating f at the $n + 1$ equally spaced nodes in $[-1, 1]$, including -1 and 1 , then

$$\lim_{n \rightarrow \infty} \left(\max_{x \in [-1, 1]} |p_n(x) - f(x)| \right) = 0.$$

9. **Convergence.** Let $p_n \in \mathcal{P}_n$ be the polynomial interpolating $f(x) = \sin(2x)$ at the nodes $x_i = i/n$, for $i = 0, \dots, n$. Then

$$\lim_{n \rightarrow \infty} \left(\max_{x \in [0, 1]} |p_n(x) - f(x)| \right) = +\infty.$$

10. **Convergence estimate.** If $f \in C[a, b]$ and f_n is the **piecewise quadratic** interpolant of f based on a partition of $[a, b]$ into n equally large subintervals, then

$$\lim_{n \rightarrow \infty} \left(\max_{a \leq x \leq b} |f(x) - f_n(x)| \right) = 0.$$

11. **Chebyshev polynomials.** Fix $n \in \mathbb{N}$ and let $T_n: [-1, 1] \rightarrow \mathbb{R}$ be defined by $T_n(x) = \cos(n \arccos x)$. Then T_n is a polynomial of degree n .

12. **Chebyshev polynomials.** Assume that T_2 is defined as in the previous item. Then $T_2(x) = 2x^2 - 3$. It may be useful to recall that $\cos(2x) = 2\cos(x)^2 - 1$.

13. **Chebyshev polynomials.** Let $x_0 < \dots < x_n$ denote the roots of the Chebyshev polynomial T_{n+1} on $[-1, 1]$. Using these points as interpolation nodes instead of equally spaced nodes can help reduce the maximum interpolation error.

14. **Julia.** Let $\mathbf{x} = [0, 1, 2]$ and $\mathbf{y} = [1, 2, 5]$. The following Julia code uses the Gregory–Newton approach to compute the interpolating polynomial through the three data points $(0, 1)$, $(1, 2)$, and $(2, 5)$:

```
function p(z)
    return y[1] + (y[2] - y[1]) * z
           + 1/2 * (y[3] - 2y[2] + y[1]) * z * (z + 1)
end
```

15. **Julia.** Let $\mathbf{x} = [0, 1, 2]$ and $\mathbf{y} = [1, 2, 5]$. The following Julia code correctly evaluates the Lagrange interpolating polynomial through these three data points at a point z :

```
function p(z)
    return (y[1] * (z - x[2]) * (z - x[3]) / ((x[1] - x[2]) * (x[1] - x[3]))
           + y[2] * (z - x[1]) * (z - x[3]) / ((x[2] - x[1]) * (x[2] - x[3]))
           + y[3] * (z - x[1]) * (z - x[2]) / ((x[3] - x[1]) * (x[3] - x[2])))
end
```

16. **Bonus.** Let $f \in C^\infty[a, b]$, and let f_n be the **piecewise linear** interpolant of f on a uniform mesh with mesh size $h_n = \frac{b-a}{n}$. Then

$$\lim_{n \rightarrow \infty} n \times \left(\max_{x \in [a, b]} |f(x) - f_n(x)| \right) = 0.$$

Justify briefly.